

```

if(!require(dplyr)) install.packages("dplyr")

## Loading required package: dplyr
##
## Attaching package: 'dplyr'
##
## The following objects are masked from 'package:stats':
##
##     filter, lag
##
## The following objects are masked from 'package:base':
##
##     intersect, setdiff, setequal, union
library(dplyr)

```

Assignment_1

1. Download a dataset from the web

```

url<-"https://raw.githubusercontent.com/Opensourcefordatascience/Data-sets/refs/heads/master/blood_pressure/patientsdata"
patientsdata<-read.csv(url)

```

2. Check the data import

```

head(patientsdata)

##   patient  sex agegrp bp_before bp_after
## 1      1  Male  30-45      143      153
## 2      2  Male  30-45      163      170
## 3      3  Male  30-45      153      168
## 4      4  Male  30-45      153      142
## 5      5  Male  30-45      146      141
## 6      6  Male  30-45      150      147

str(patientsdata)

## 'data.frame':   120 obs. of  5 variables:
##  $ patient  : int  1 2 3 4 5 6 7 8 9 10 ...
##  $ sex      : chr  "Male" "Male" "Male" "Male" ...
##  $ agegrp   : chr  "30-45" "30-45" "30-45" "30-45" ...
##  $ bp_before: int  143 163 153 153 146 150 148 153 153 158 ...
##  $ bp_after : int  153 170 168 142 141 147 133 141 131 125 ...

summary(patientsdata)

```

```

##      patient          sex          agegrp          bp_before
##  Min.   : 1.00   Length:120   Length:120   Min.     :138.0
##  1st Qu.: 30.75   Class :character   Class :character   1st Qu.:147.0
##  Median : 60.50   Mode  :character   Mode  :character   Median :154.5
##  Mean   : 60.50                                     Mean   :156.4
##  3rd Qu.: 90.25                                     3rd Qu.:164.0
##  Max.   :120.00                                     Max.   :185.0
##      bp_after
##  Min.     :125.0
##  1st Qu.:140.8
##  Median :149.5
##  Mean    :151.4

```

```
## 3rd Qu.:161.0
## Max. :185.0
```

3. Descriptive statistics

```
summary(patientsdata$bp_before)
```

```
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##    138.0  147.0   154.5   156.4   164.0   185.0
```

```
summary(patientsdata$bp_after)
```

```
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##    125.0  140.8   149.5   151.4   161.0   185.0
```

```
table(patientsdata$sex)
```

```
##
## Female    Male
##      60     60
```

```
table(patientsdata$agegrp)
```

```
##
## 30-45 46-59  60+
##     40   40   40
```

4. Transforming variables

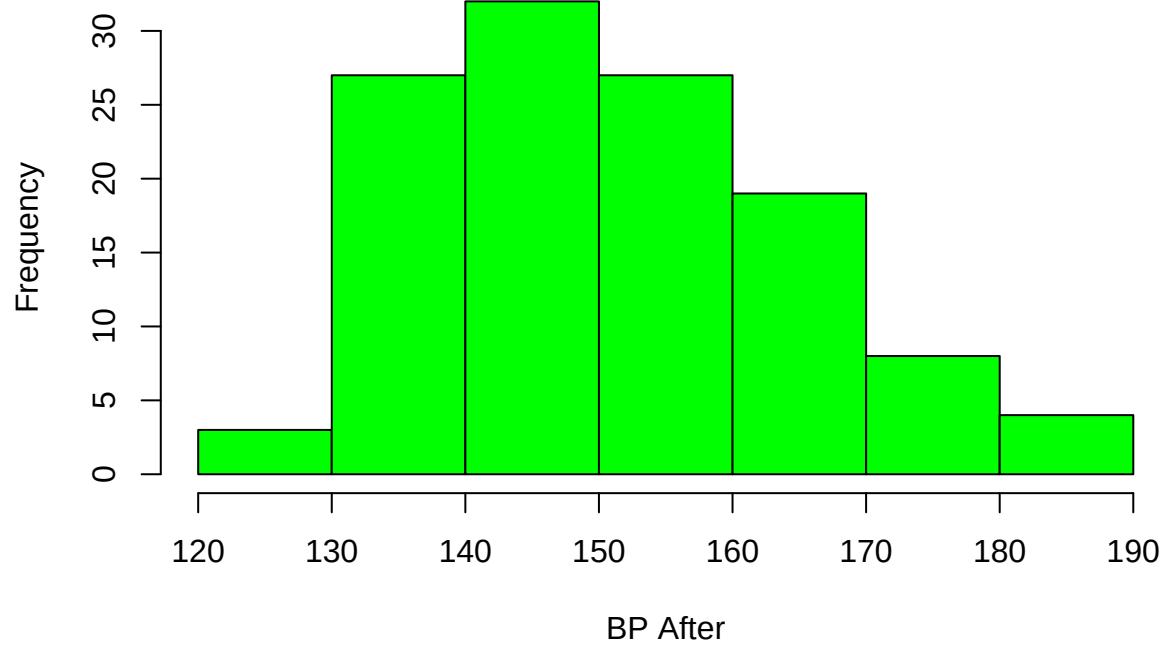
```
patientsdata<-patientsdata %>%
  mutate(agegrp_2=recode(agegrp, "30-45"="Young", "46-59"="Middle Age", "60+"="Seniors"))
table(patientsdata$agegrp_2)
```

```
##
## Middle Age    Seniors      Young
##          40         40         40
```

5. Charts Histogram

```
hist(patientsdata$bp_after,
      main="Histogram of Patients BP after",
      xlab="BP After",
      col="green")
```

Histogram of Patients BP after



Scatter plot

```
plot(patientsdata$bp_before,patientsdata$bp_after,  
      main="Scatter plot of BP Before Vs BP After",  
      ylab="BP After",  
      xlab="BP Before",  
      col="blue",  
      pch=15  
)
```

Scatter plot of BP Before Vs BP After

